ENTERPRISE / SOFTWARE

LIVE

Systems Monitoring Dashboard

Real-time infra dashboard with sub-second latency, WebSocket streaming, time-series aggregation, and anomaly alerts.

`React` `WebSocket` `InfluxDB` `Python`

VERSION	STATUS	LICENSE	INGEST	OWNER	UPDATED
3.1.0	Live	Internal	1.2M pts/s	Observability	Sep 2026

1. Overview

The Systems Monitoring Dashboard renders live infrastructure state with sub-second lag between an agent reading a counter and an operator seeing it move. A Python ingest tier accepts metrics over gRPC and line protocol, writes them to InfluxDB, and simultaneously fans the same points out to subscribed browsers over WebSocket, so the display path never waits on a storage round trip.

The dashboard is built for incident work rather than reporting. Panels are defined in JSON and composed into boards; every panel shares one subscription multiplexer, so a board with forty panels opens one socket rather than forty. Anomaly scoring runs continuously on the server side and annotates the stream inline, which means an operator sees the flagged deviation on the same chart that carries the raw series.

2. Data path

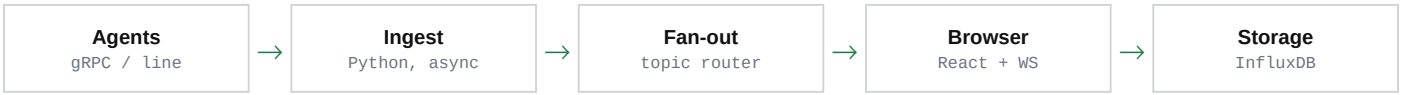

Figure 1 — The live path and the durable path diverge at ingest; a storage stall degrades history and alert backfill, never the live view.

Ingest workers batch writes on a 200 ms window while forwarding individual points to the router immediately. The router keys subscriptions by `measurement + tag set` and applies server-side decimation: a panel showing a one-hour window receives pre-aggregated buckets, not every raw point, which keeps socket bandwidth flat as cardinality grows.

3. Component layers

Collection	Agent SDKs plus a Prometheus scrape shim. At-least-once delivery with a local spool for network gaps.
Ingest	Async Python workers: parse, validate, tag-normalise, batch to storage, publish to the router.
Router	In-memory topic index with per-connection backpressure; slow clients are decimated, never blocking.
Storage	InfluxDB with tiered retention and continuous downsampling tasks.
Client	React with a virtualised canvas renderer; one multiplexed socket per board.

4. Retention and aggregation

TIER	RESOLUTION	RETENTION	SOURCE
Hot	1 s	6 hours	Raw writes from ingest.
Warm	10 s	14 days	Continuous downsample task.
Cold	1 min	13 months	Rollup with min/max/mean/p95 preserved.

5. Subscription protocol

```
// client -> server
{ "op": "subscribe",
  "id": "panel-7",
  "query": {
    "measurement": "node_cpu_seconds",
    "tags":  { "cluster": "prod-us-east", "mode": "user" },
    "window": "1h",
    "agg":    "mean",        // mean | max | p95 | rate
    "step":   "5s"           // server decimates to this resolution
  },
  "anomaly": { "detector": "ewma", "sensitivity": 0.85 } }

// server -> client (frame, ~5 Hz)
{ "id": "panel-7", "t": 1789012345678,
  "points": [[1789012345, 0.41], [1789012350, 0.44]],
  "flags":  [{ "t": 1789012350, "score": 0.93, "kind": "level_shift" }] }
```

Listing 1 — Backfill for the requested window arrives as one snapshot frame, then deltas stream.

6. Anomaly detection

- **Detectors.** Three run server-side per subscribed series: EWMA residual for level shifts, seasonal decomposition for daily and weekly cycles, and a rate-of-change guard for cliffs. Each emits a normalised score between 0 and 1.
- **Alerting.** A flag becomes an alert only after it persists for the configured dwell time and clears a cluster-wide correlation check, which suppresses the fan-out of a single noisy upstream into hundreds of dependent alerts.
- **Tuning.** Sensitivity is per panel and takes effect on the live stream immediately; operators can replay the last 24 hours against a candidate setting before saving it, so threshold changes are evaluated rather than guessed.

7. Measured performance

DIMENSION	MEASURED	NOTES
Ingest throughput	1.2M points/s	Six workers, 4 vCPU each, batched writes.
Agent-to-glass latency	310 ms p50 / 780 ms p95	Includes network, routing, and render.
Concurrent boards	2,400 sockets	Per router node; scales horizontally by topic shard.
Active series	8.5M	Cardinality ceiling before write amplification degrades.

8. Operations

- Router nodes are stateless apart from subscription state, which clients re-establish on reconnect with an exponential backoff and a resume token that avoids refetching the window.
- The ingest tier sheds load by dropping the live fan-out first and preserving durable writes, on the principle that a stale chart is recoverable and a gap in history is not.
- Capacity planning tracks active series rather than points per second; cardinality growth from unbounded tag values is the most common cause of degradation and is alerted on directly.

9. Roadmap

RELEASE	PLANNED WORK	TARGET
3.2	Exemplar links from a flagged point to the matching distributed trace.	Q4 2026
3.3	Board-level templating with cluster and namespace variables.	Q1 2027
4.0	Pluggable storage backend; Prometheus remote-read as a first-class source.	Q2 2027